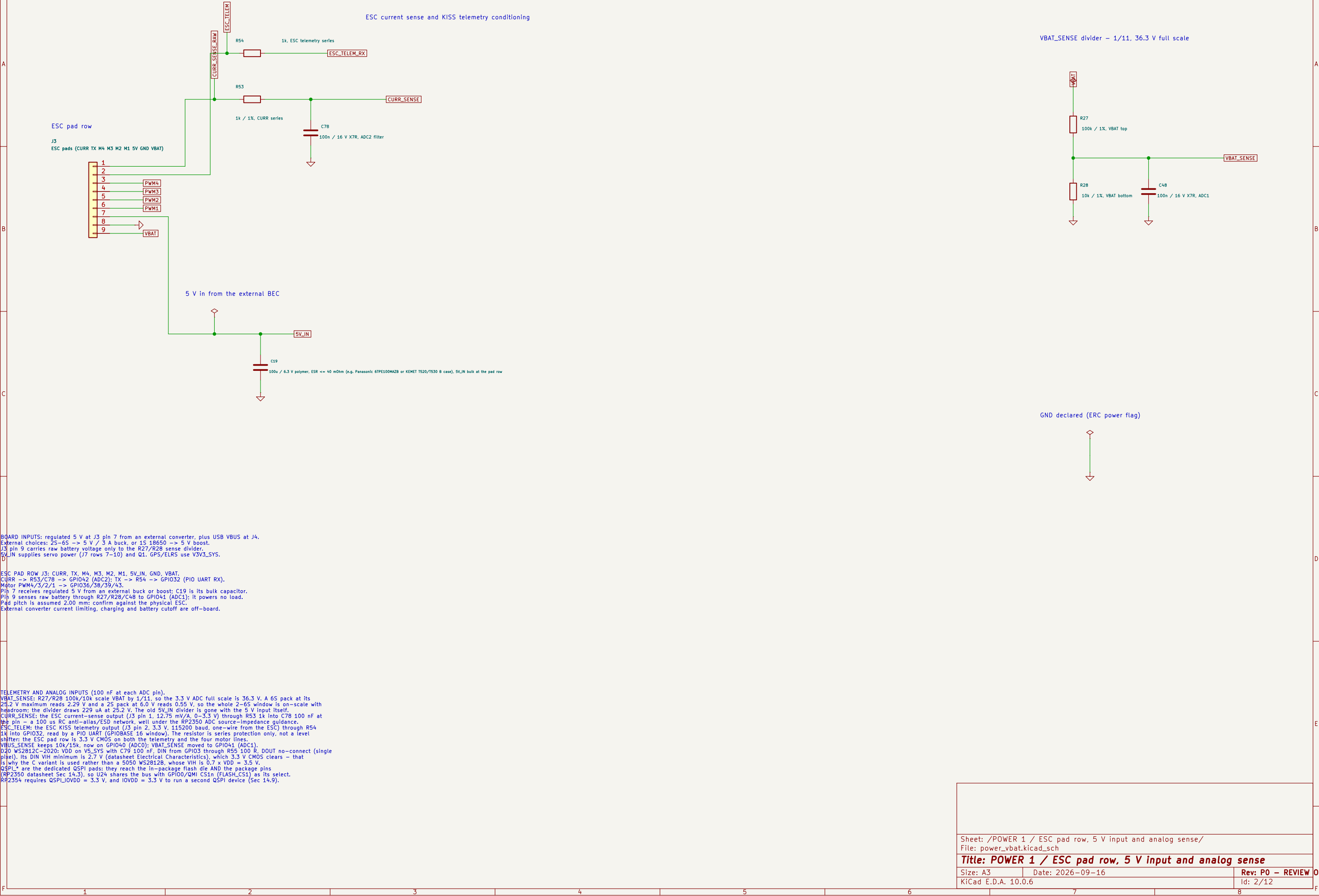


	1	2	3	4	5	6	7	8
A	MARV V2 flight controller – schematic index. One page per subsystem; on each page the IC is drawn with its own decoupling, pull–ups, series resistors, test points and connectors wired to it, and a global label is used only where a net leaves the page. See DESIGN_SPEC.md for the design itself. Inputs: regulated 5 V from an EXTERNAL BEC on the MicoAir AM32 ESC pad row J3 pin 7, and USB VBUS. J3 pin 9 brings raw pack VBAT in as a SENSE wire only – no load, no bulk, no copper zone. Rails: 5V_IN is the external converter rail and servo bus; V5_SYS is 5V_IN OR USB through the Q1/D1 OR and feeds only the two 3.3 V regulators and the status LED; V3V3_SYS from the AP63203 buck, V3V3_ANA from the TPS7A20 LDO.							
B	POWER 1 / ESC pad row, 5 V input and analog sense File: power_vbat.kicad_sch		BOARD 5 / USB–C, D+/D– series pair, SWD pads File: usb_debug.kicad_sch		BOARD 3 / microSD socket File: sd.kicad_sch			
C	POWER 2 / 5 V OR – BEC P–FET and USB Schottky File: power_or.kicad_sch		BOARD 2 / ICM–45686 six–axis IMU File: imu.kicad_sch		BOARD 3 / optional QSPI expansion socket (DNP) File: qspi_expansion.kicad_sch			
D	POWER 3 / 3.3 V system buck and analog LDO File: power_3v3.kicad_sch		BOARD 2 / BMP581 barometer File: baro.kicad_sch		BOARD 4 / IO array, status LED, mounting holes File: io_block.kicad_sch			
E	BOARD 1 / RP2354B File: mcu.kicad_sch		BOARD 2 / ADXL375 high–g accelerometer File: highg.kicad_sch					
F								Sheet: / File: MARV–V2.kicad_sch Title: MARV–V2 – schematic index Size: A3Date: 2026–09–16KiCad E.D.A. 10.0.6 Rev: P0 – NOT FOR FABId: 1/12
	1	2	3	4	5	6	7	8

POWER 1 – ESC PAD ROW J3 (NINE PADS), 5 V INPUT FROM AN EXTERNAL BEC, VBAT SENSE, ANALOG SENSE.
Regulated 5 V arrives on J3 pin 7 as 5V_IN; pin 9 brings raw pack VBAT in as a SENSE wire only.



POWER 2 – Q1 A03401A P–FET + D1 1N5819WS SCHOTTKY. 5V_IN (external BEC, J3 pin 7) OR USB_VBUS –> V5_SYS.
USB present turns Q1 OFF, so a host can never back-feed the BEC, the servo rows or the ESC.

A

B

C

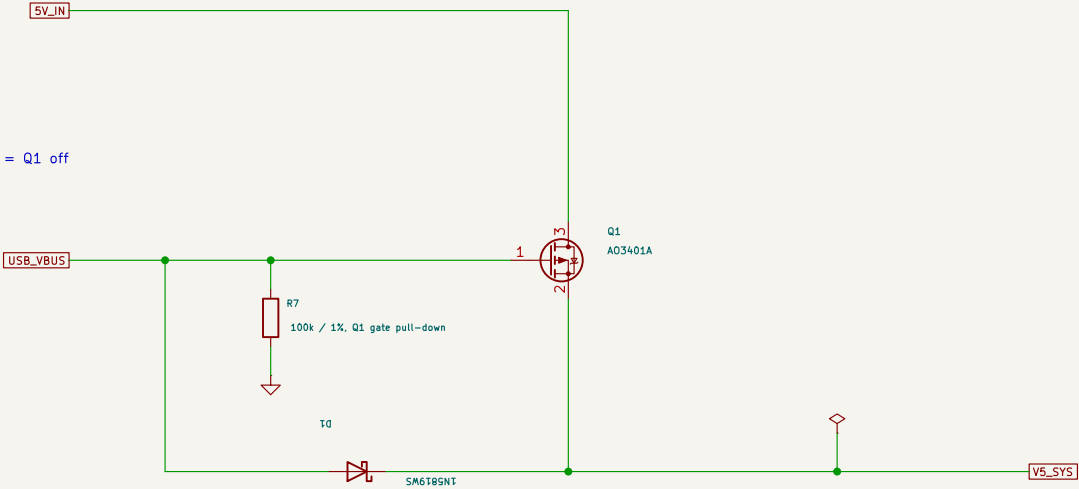
D

E

F

BEC path – Q1 P–channel FET, drain 5V_IN, source V5_SYS

Gate on USB_VBUS – USB present = Q1 off



USB path – D1 Schottky, the only way USB reaches V5_SYS

V5_SYS – the two 3.3 V regulators and the status LED, nothing else

5 V INPUT OR: Q1 A03401A and D1 1N5819WS.
Q1 drain = 5V_IN, source = V5_SYS, gate = USB_VBUS. R7 pulls the gate down.
D1 anode = USB_VBUS; cathode = V5_SYS.
Without USB, Q1's channel carries the external supply after its gate discharges.
With USB present, Q1 can be off; D1 and Q1's body diode can share the load.
There is no guaranteed BEC priority or lossless switchover.
C19 (100 uF) is on 5V_IN; C8/C25/C79 are on V5_SYS.
V5_SYS feeds U7 (system buck), U12 (sensor LDO) and the RGB status LED.
GPS/ELRS and other 3.3 V I/O use U7's output; servos stay on external 5V_IN.
Input-only simulation: simulations/power_input.cir. See DESIGN_SPEC.md.

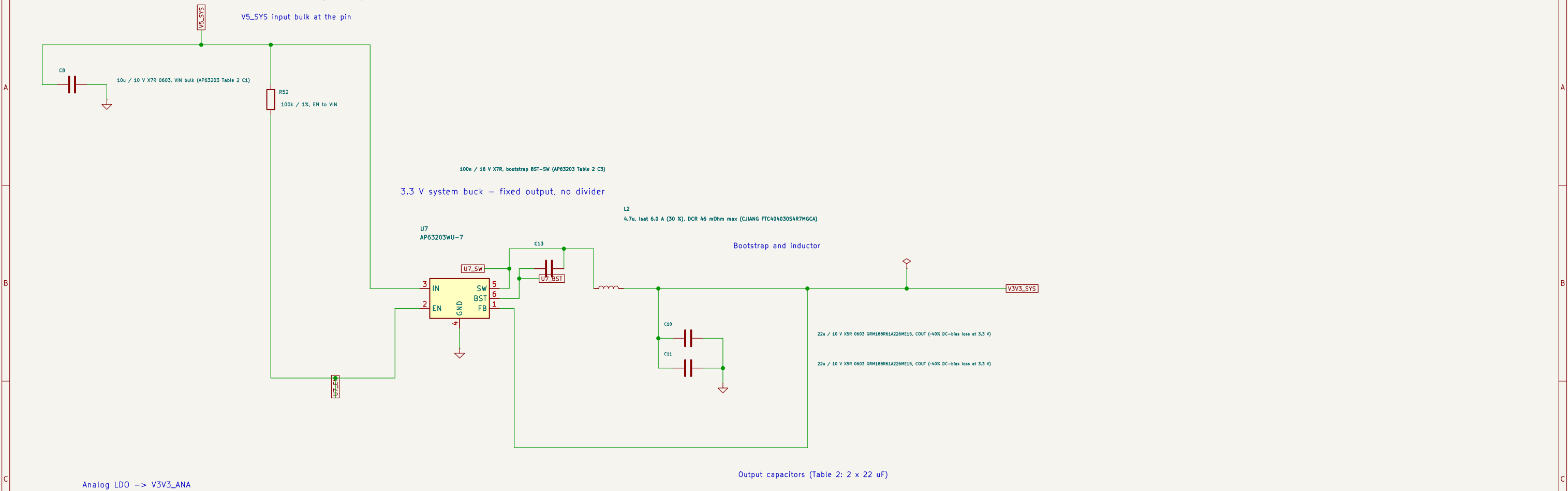
USB_VBUS feeds the D1 Schottky and the Q1 gate on POWER 2 directly; no inrush or current limiting on this sheet, and no data ESD array any more – U8 USBLC6–2SC6 was JLC Extended and is removed (DESIGN_SPEC "Decisions"), so D+/D– go straight from J4 into the 27 R series pair R22/R23, as on the madflight FC3v2. The RP2354B USB pins carry their own on–die ESD structures; a board–level array is what this design gives up, deliberately.

Sheet: /POWER 2 / 5 V OR – BEC P–FET and USB Schottky/ File: power_or.kicad_sch			
Title: POWER 2 / 5 V OR – BEC P–FET and USB Schottky			
Size: A3	Date: 2026–09–16	Rev: P0 – REVIEW ONLY	
KiCad E.D.A. 10.0.6	Id: 3/12		

F

POWER 3 – U7 AP63203WU-7 FIXED 3.3 V BUCK (V5_SYS -> V3V3_SYS) AND U12 TPS7A20 LDO (-> V3V3_ANA)

AVIONICS ONLY: 300 mA continuous / 500 mA short peak, provisional. NO SERVO POWER.



C
Analog LDO -> V3V3_ANA

U12
TP5A2033PDBVR

V5_SYS

C25

1u / 10 V X7R 0402, LDO Input

1
3
5
2

IN
EN
OUT
GND

V3V3_SYS

C26

2.2u / 16 V X5R 0603, LDO output (>=1 uF effective, SBVS338H Sec 7.1.1)

V3V3_ANA

AD65203WU-7, VS_5V5 —> V3V3_V5S BUCK (Diodes DS541326 Rev. 3-2), 3.8–3.2 V in, FIXED 3.3 V out, 2 A,
1.1 MHz with +/−6% frequency spread spectrum, 4 ms internal soft-start, 22 uA quiescent, TS026E.
Replaces the AD652913 two-stage design. The new FB1 has a 0.1 μ F divider + five 22 uF at -8 parts instead of -25.
PARTS LIST ARE TABLE 2. Recommended Component Selections for AP65203) (3.3 V output) and Figure 21, the AP65203/AP63205
typical application, with one documented substitution:
C1, 3.9 uF in Table 2 —> 4.7 uF (L2). Sec 10 "Inductor" licenses it explicitly: "It is recommended to select an
inductor of approximately 2.2uH to 10uH with a DC current rating of at least 3x higher than the maximum load
current." Using a larger inductance for improved efficiency under light load conditions," and this rail is at
a light load (~180 mA typical / ~300 mA peak), 4.7 uH is a part the board already sources.
C10 10 uF —> CB, 10 uF / 25 V XSR 0603, at VIN, and it is the ONLY input capacitor; the C9 100 nF HF bypass
that used to sit beside it was an addition beyond Table 2 and is removed.
C2, 2 x 22 uF —> C40 / C41 1uF / 40 V XSR 0603 (GRM18R61A226ME15) at the output, excepting the table value,
C3 100 nF —> C13, bootstrap BST—SW.
FB (pin 1) is a SENSE input on the fixed-output parts and goes straight to V3V3_V5S — Sec 9 "Setting the Output
Voltage": only the adjustable AP65203/AP63201 take a divider (Eq.6). So there is no feedback divider, no 0.1 μ F
input cap, and the divider is not needed. Note that the output is the same as the input: VOUT is the same as VIN.
VIN (pin 2): "The en pin is a high voltage pin and can be directly connected to VIN", abs max 35 V, threshold
1.8 V rising / 1.0 V falling, internal 1.5 uA pull-up, RS2 100k from VS_5V5, is that connection with a series
element, so a UVLO divider (Eq.1, Eq.2) or a start-delay cap (Eq.3) can be added later without touching copper.
If the divider is used, the divider ratio = 3.3 V, VOUT / 1.1 V, 3.0 = 3.3, 2.2 μ F, 0.22 A, 0.22 A, 0.22 A, 0.22 A,
pk-pk (0.27 A at the −20% inductance limit), so D8 gives a 0.41 inductor peak at the 0.3 A peak rail load.
On USB, VIN falls to −4.5 V behind D1 and dIL drops to 0.17 A; the part starts in CCM—adjusted light-load operation
either way and neither case comes near the 2 A rating or the inductor.
If the divider is not used, the divider ratio = 3.3 V, VOUT / 1.1 V, 3.0 = 3.3, 2.2 μ F, 0.22 A, 0.22 A, 0.22 A, 0.22 A,
pk-pk (0.27 A at the −20% inductance limit), so D8 gives a 0.41 inductor peak at the 0.3 A peak rail load.
On USB, VIN falls to −4.5 V behind D1 and dIL drops to 0.17 A; the part starts in CCM—adjusted light-load operation
either way and neither case comes near the 2 A rating or the inductor.
If the divider is not used, the divider ratio = 3.3 V, VOUT / 1.1 V, 3.0 = 3.3, 2.2 μ F, 0.22 A, 0.22 A, 0.22 A, 0.22 A,
pk-pk (0.27 A at the −20% inductance limit), so D8 gives a 0.41 inductor peak at the 0.3 A peak rail load.
On USB, VIN falls to −4.5 V behind D1 and dIL drops to 0.17 A; the part starts in CCM—adjusted light-load operation
either way and neither case comes near the 2 A rating or the inductor.

C2, 2 x 22 uF the table asks for in nominal terms and is the same derating the removed stage carried.
C2, CIJANG FC4040305AR7MGCC (LSCD C396760B3), the inductor the removed AP65205 stage used as L3 – 4.7 uH +/-20%,
0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A,
molded metal core, shielded (CSZ CIJANG FIC series datasheet Rev 7.0 2025/11/05, Notes 3/4/7); the usage rating
is the LESSER of IRMS and ISAT, so 4.0 A. At a 0.41 A peak that is a 10x margin – enormously over-rated for this
rail, and deliberately so: it is a part already on the BOM and on the reel, and it costs 4¢ mNm x 0.3 A^2 = 4¢ mNm.
If the divider is not used, the divider ratio = 3.3 V, VOUT / 1.1 V, 3.0 = 3.3, 2.2 μ F, 0.22 A, 0.22 A, 0.22 A, 0.22 A, 0.22 A,
pk-pk (0.27 A at the −20% inductance limit), so D8 gives a 0.41 inductor peak at the 0.3 A peak rail load.
On USB, VIN falls to −4.5 V behind D1 and dIL drops to 0.17 A; the part starts in CCM—adjusted light-load operation
either way and neither case comes near the 2 A rating or the inductor.

The same library footprint family and 2.2 uH is still inside the dataset's 2.2–10 uH window – at 0.74 A pk-pk ripple.

SV3A_AVD0 feeds ICM-45686, BMP581, ADXL375 and the RP235A4B ADC_AVD0 pin (100 nF at each pin on the MCU sheet); VREG_AVD0 is filtered from SV3V_SYS instead, so the analog rail carries no core-regulator current. U12 (en pin '3') is tied to SV3V_SYS

The AP63203 has no power-good output, so the sequencing is done by the rail EN rather than by a status pin – the LDO enables

when the buck output is up and drops with it, so the sequencing is the same buck-before-LDO ordering PWR_GOOD used to give, at no parts.

What is lost is the "in regulation" qualifier: EN now releases at U12's own -0.9 V enable threshold on a rising SV3V_SYS.

Not at the point a comparator declares the buck settled. For a 10–20 mA sensor rail behind a 4 ms soft-start that is a factor of a few microseconds of difference, and the sensors are helped by firmware for far longer than that.

C25 / C26, The LDO's OWN CAPACITORS (TPS7A20 datasheet SBV5338H): the part is "stable with 1 uF of minimum ceramic output capacitor", and Sec 7.1.1 says the recommended values already assume "the effective capacitance to decrease by as much as 50%" – i.e. the 1 uF is an EFFECTIVE floor, not a nameplate one. C26 is therefore

16 V 15 V X5R in an 0603 (LCSC C26350, J1848), and the 1 uF 0402 it was used at 3.3 V

has no headroom left against that floor, and the 16 V 0603 keeps well over 1 uF at 3.3 V. C25 stays 1 uF 0402 at the input, where the datasheet asks only for "at least equal to the nominal value" and nothing depends on it.

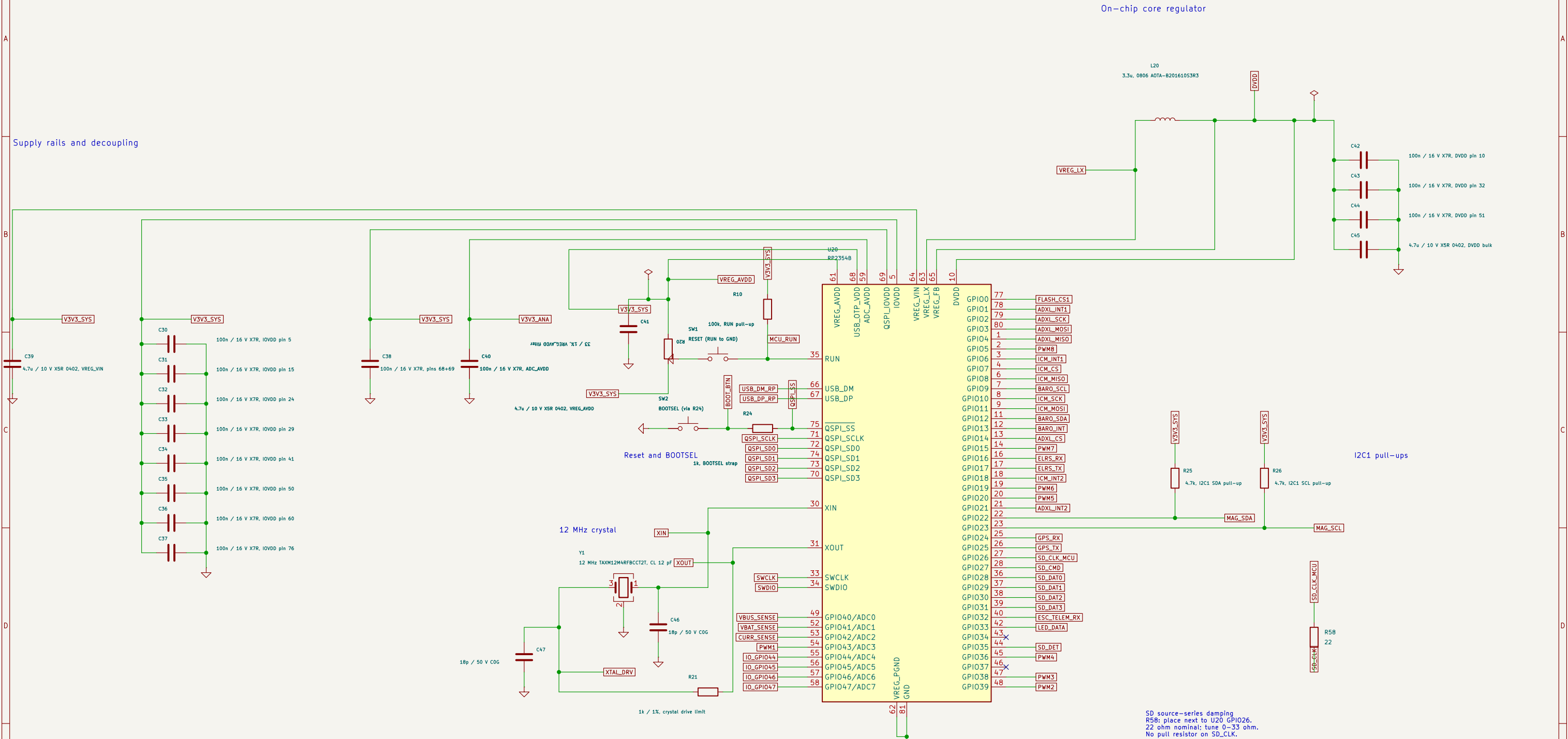
PASSIVE PACKAGES (DESIGN_SPEC "Physical design"): resistors are 0402, ceramics ≤ 4.7 uF are 0402 and the 10-22 uF ceramics are 0603. RESISTORS ARE 0402. NOT 0201: JLCPCB has no Basic 0201 resistor of any value, so every 0201 line carried a per-unique-part-Extended setup fee. see reports/JLC-audit.md, 0402 makes most of the difference in the hand-reworkable vs. not-reworkable. THE 0.4 uF PAIR IS ONE WITH THE TP562913; the AP63203 is a fixed-output part and has no feedback divider, so no 0.1 μ thin-film line is ordered at all. C10/C11 (AP63203 COUT) are 22 μ / 10 μ XSR 0603 (GRM188R61A226M15). At 3.3 V DC bias an XSR 22 μ 0603 keeps roughly 60% of its nominal value, so the 44 μ of nominal is -26 uF effective. Do not read the effective value off the BOM. The 22 μ 0603 ceramics remain machine-in-place parts - JLC assembles them, a bench iron will not rework and C19, THE 5VJN bulk, IS 100 μ / 6.3 V POLYMER IN AN EIA-3528-21 (B) CASE. It moved from V5_S5YS to 5V_JVN with the BEC decision: hold-up belongs on the rail the servo and module runs on, which is now a half of the OR-ing FET, and V5_S5YS keeps only C8 at the AP63203 input. ESR must stay ≤ 40 mOhm: Panasonic 6TPE100MAZB or a KEMET T520/T530 B case, 6.3 on a 5.0 uF rail line is a 1.26x derating, the normal polymeric figure; a tantalum electrolytic is not used. THE GPSSB DECKS IN simulations/ STILL MODEL THE OLD TOPOLOGY (a TP52121 mux fed by an on-board pack buck, and a TP562913-class 3.3 V stage). They were not re-derived for the BEC + P-FET + AP63203 tree, so every number they produce is now history, not evidence - see DESIGN_SPEC "Open items".

Sheet: /POWER 3 / 3.3 V system buck and analog LDO/
File: power_3v3.kicad_sch

Title: POWER 3 / 3.3 V system buck and analog LDO

Size: A3	Date: 2026-09-16	Rev: P0 – REVIEW	ONLY
KiCad E.D.A. 10.0.6		Id: 4/12	

BOARD 1 – RP2354B, QFN–80, 2 MB IN–PACKAGE QSPI FLASH. Digital rails on V3V3_SYS, core DVDD from the on–chip buck. Supply pins fan out on the left into one decoupling group per rail.



GPIO MAP (pin = QFN–80 pin)	
GPIO0	77 FLASH_CS1 QM1 CS1n, log flash select
GPIO1	78 ADXL_INT1 ADXL375 INT1
GPIO2	79 ADXL_SCK SPI0 SCK
GPIO3	80 ADXL_MOSI SPI0 TX
GPIO4	1 ADXL_MISO SPI0 RX
GPIO5	2 PWM5 PWM2 B -> J6 row 10 (S8)
GPIO6	3 ICM_INT1 ICM-45686 INT1
GPIO7	4 ICM_CS ICM-45686 CS
GPIO8	6 ICM_MISO SPI1 RX
GPIO9	7 BARO_SCL I2C0 SCL
GPIO10	8 ICM_SCK SPI1 SCK
GPIO11	9 ICM_MOSI SPI1 TX
GPIO12	11 BARO_SDA I2C0 SDA
GPIO13	12 BARO_INT BMP581 INT
GPIO14	13 ADXL_CS ADXL375 CS
GPIO15	14 PWM7 PWM7 B -> J6 row 9 (S7)
GPIO16	16 ELRS_RX UART0 TX -> radio RX
GPIO17	17 ELRS_TX UART0 RX -> radio TX
GPIO18	18 ICM_INT2 ICM-45686 INT2
GPIO19	19 PWM6 PWM1 B -> J6 row 8 (S6)
GPIO20	20 PWM5 PWM2 A -> J6 row 7 (S5)
GPIO21	21 ADXL_INT2 ADXL375 INT2
GPIO22	22 MAG_SDA I2C1 SDA
GPIO23	23 MAG_SCL I2C1 SCL

RP2354B, QFN–80, 2 MB in–package QSPI flash. Digital rails on V3V3_SYS, core DVDD from the on–chip buck. RAIL SPLIT: V3V3_ANA (TPS7A20 LDO) now feeds exactly one RP2354B pin – ADC_AVDD (pin 59, with C40). The core regulator input flows out filter R20/C41 that produces VREG_AVDD is fed from V3V3_SYS instead, so the buck–converter reference current no longer of the low–noise analog rail and the sensors do not share a rail with it; V3V3_ANA is left carrying only ADC_AVDD and the MEMS. Decoupling, core regulator, crystal and BOOTSEL follow the RP2350 hardware design guide; every pin net is a global label.

PORT NAMING: a port label is named for the peripheral pin it comes from, so GPS_TX / ELRS_TX are module outputs and land on MCU UART RX pins, while GPS_RX / ELRS_RX are module inputs fed by the MCU UART TX pins. On the IO array (POWER 3) the UART rows are therefore 1 GPS_RX (silK T0) / 2 GPS_TX (R0) / 3 ELRS_RX (T1) / 4 ELRS_TX (R1), rows 5/6 are MAG_SDA / MAG_SCL rows 7–10 the PWM5–8 servo signals and rows 11–14 the four exposed spares. The array is three stacked 1x14 rows: J6 SIGNAL, J7 POWER, J8 GND. All array SIGNAL pins are 3.3 V CMOS. GPIO37 (pin 46) is SPARE and unconnected again. It used to read PWR_SRC_ST, the open–drain status output of the U25 TPS2121 mux, and the mux is gone. USB presence is already readable on the VBUS_SENSE divider (GPIO40).

SPARE IO: GPIO44–47 remain on IO block J6 rows 11–14. GPIO34 and GPIO37 are unexposed and carry no–connect flags. GPIO budget: 42 assigned, 4 exposed spares, 2 unconnected. SPI0: ADXL375 only. SPI1: ICM–45686 only. I2C0: BMP581 only. Native SD: GPIO26–31, PIO + DMA. Independent sensor interrupts. See PINOUT.md for the complete allocation.

Crystal circuit per design guide Sec 4, Y1 IS TAXM12M4RFBCT2T (LCSC C133337), 12 MHz, SMD 3225 4–pad, CL = 12 pF – the ABM8–272–T3 the earlier revision named is not orderable at JLCPCB in any form (reports/Jlc–audit.md section 1). LOAD CAPS FOLLOW THE CRYSTAL, NOT THE OTHER WAY ROUND: CL = C46+C47/(C46+C47) + Cstray, so C46/C47 are 18 pF COG each (2 x 18 pF in series = 9 pF, plus ~3 pF of XIN/XTAL_DRV pin and trace stray = 12 pF, matching CL exactly). They were 15 pF against the old CL 10 pF part; leaving 15 pF on a CL 12 pF crystal gives 10.5 pF of load, i.e. an under–loaded oscillator running fast by roughly the trim sensitivity of the part. The ~3 pF stray is the design–guide estimate and is the one term here that is not a datasheet number – if the board is ever pulled fbr frequency, that is the term to measure. 1 k in the XOUT leg to limit drive with a 50 ohm ESR crystal. C38 is the single design–guide decoupling exception

(pins 68+69 share it). VREG_AVDD RC = 33 ohm + 4.7 uF (design guide Sec 2.1), taken from V3V3_SYS; VREG_FB is tied to DVDD, VREG_PGND to GND, EP to GND.

SD source–series damping. R58: place next to U20 GPIO26. 22 ohm nominal; tune 0–33 ohm. No pull resistor on SD_CLK.

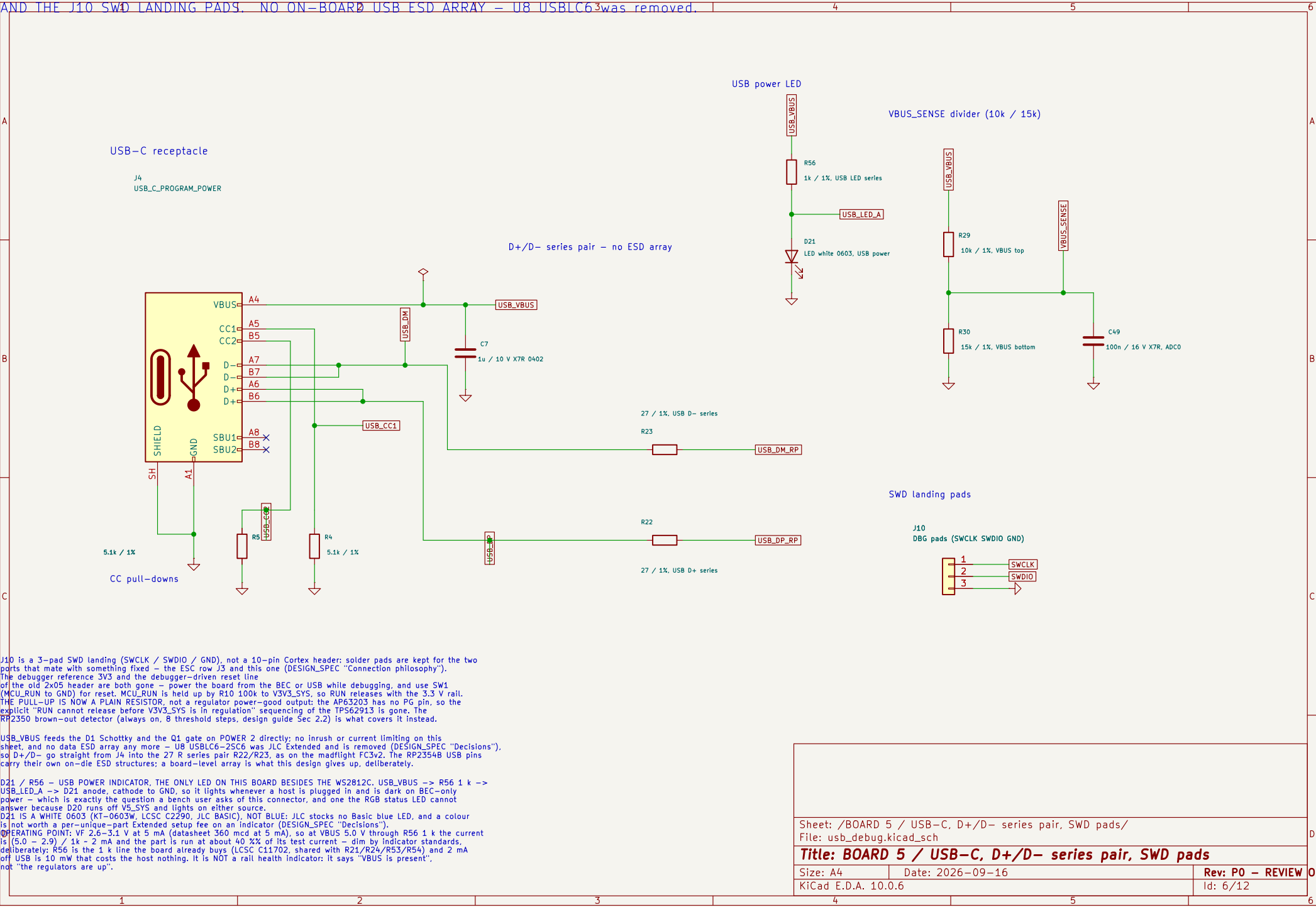
Sheet: /BOARD 1 / RP2354B/
File: mcu.kicad_sch

Title: BOARD 1 / RP2354B

Size: A3 Date: 2026–09–16
KiCad E.D.A. 10.0.6

Rev: P0 – REVIEW ONLY
Id: 5/12

BOARD 5 – J4 USB-C 2.0 RECEPTACLE, THE 27 R D+/D- SERIES PAIR, THE VBUS SENSE DIVIDER AND THE J10 SWD LANDING PADS. NO ON-BOARD USB ESD ARRAY – U8 USBLC6³ was removed.



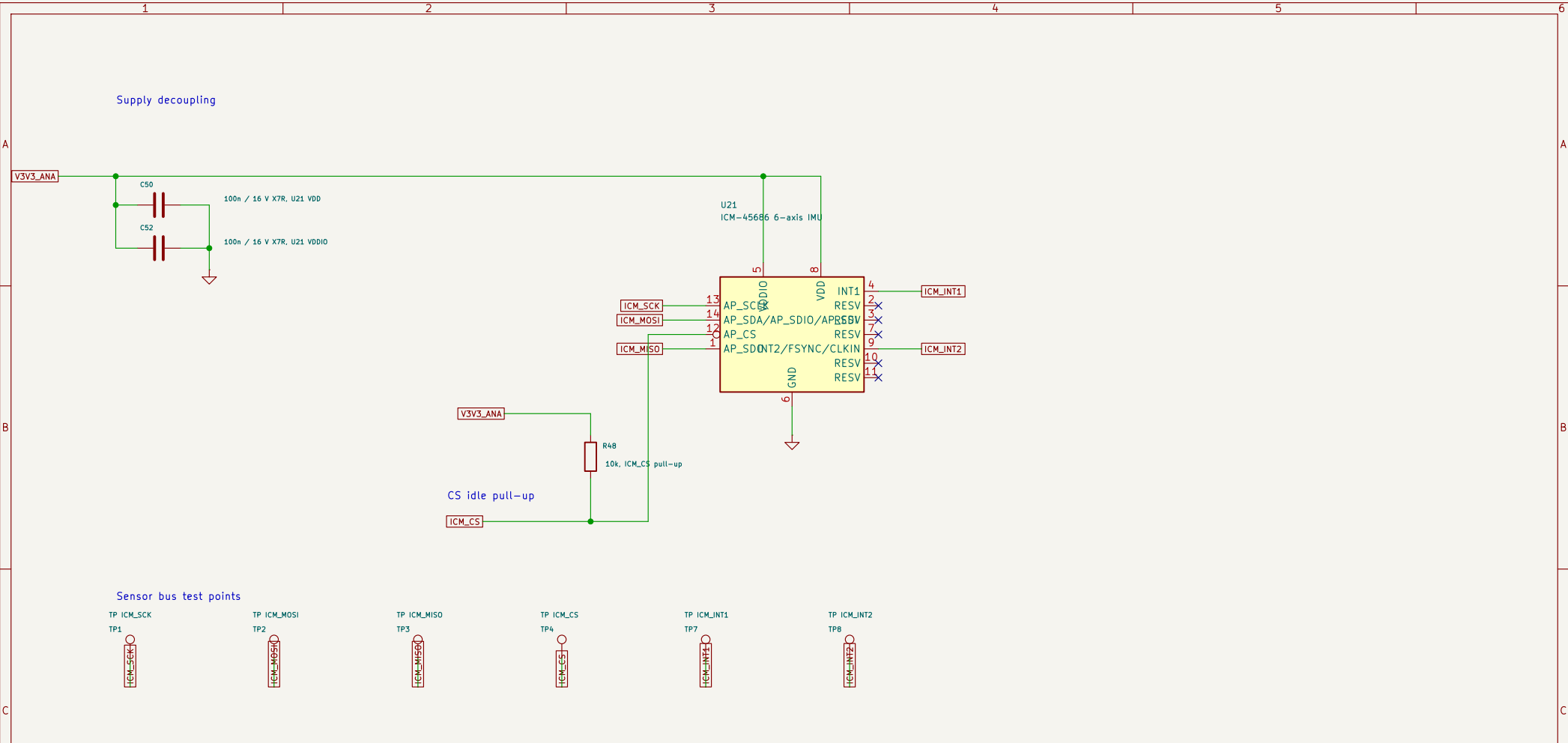
J10 is a 3-pad SWD landing (SWCLK / SWDIO / GND), not a 10-pin Cortex header: solder pads are kept for the two ports that mate with something fixed – the ESC row J3 and this one (DESIGN_SPEC “Connection philosophy”). The debugger reference 3V3 and the debugger-driven reset line of the old 2x05 header are both gone – power the board from the BEC or USB while debugging, and use SW1 (MCU_RUN to GND) for reset. MCU_RUN is held by R10 100k to V3V3_SYS, so RUN releases with the 3.3 V rail. THE PULL-UP IS NOW A PLAIN RESISTOR, not a regulator power-good output: the AP63203 has no PG pin, so the explicit “RUN cannot release before V3V3_SYS is in regulation” sequencing of the TP562913 is gone. The RP2350 brown-out detector (always on, 8 threshold steps, design guide Sec 2.2) is what covers it instead.

USB_VBUS feeds the D1 Schottky and the Q1 gate on POWER 2 directly: no inrush or current limiting on this sheet, and no data ESD array any more – U8 USBLC6–25C6 was JLC Extended and is removed (DESIGN_SPEC “Decisions”). so D+/D- go straight from J4 into the 27 R series pair R22/R23, as on the madflight FC3v2. The RP2354B USB pins carry their own on-die ESD structures: a board-level array is what this design gives up, deliberately.

D21 / R56 – USB POWER INDICATOR, THE ONLY LED ON THIS BOARD BESIDES THE WS2812C. USB_VBUS -> R56 1 k -> USB_LED_A -> D21 anode, cathode to GND, so it lights whenever a host is plugged in and is dark on BEC-only power – which is exactly the question a bench user asks of this connector, and one the RGB status LED cannot answer because D20 runs off V5_SYS and lights on either source. D21 IS A WHITE 0603 (KT-0603W, LCSC C2290, JLC BASIC), NOT BLUE: JLC stocks no Basic blue LED, and a colour is not worth a per-unique-part Extended setup fee on an indicator (DESIGN_SPEC “Decisions”). OPERATING POINT: VF 2.6–3.1 V at 5 mA (datasheet 360 mcd at 5 mA), so at VBUS 5.0 V through R56 1 k the current is (5.0 – 2.9) / 1k – 2 mA and the part is run at about 40 % of its test current – dim by indicator standards, deliberately: R56 is the 1 k line the board already buys (LCSC C11702, shared with R21/R24/R53/R54) and 2 mA off USB is 10 mW that costs the host nothing. It is NOT a rail health indicator: it says “VBUS is present”, not “the regulators are up”.

Sheet: /BOARD 5 / USB-C, D+/D- series pair, SWD pads/ File: usb_debug.kicad_sch			
Title: BOARD 5 / USB-C, D+/D- series pair, SWD pads			
Size: A4	Date: 2026-09-16	Rev: P0 – REVIEW ONLY	
KiCad E.D.A. 10.0.6			Id: 6/12

BOARD 2 – U21 ICM–45686: dedicated hardware SPI1. Powered from V3V3_ANA.



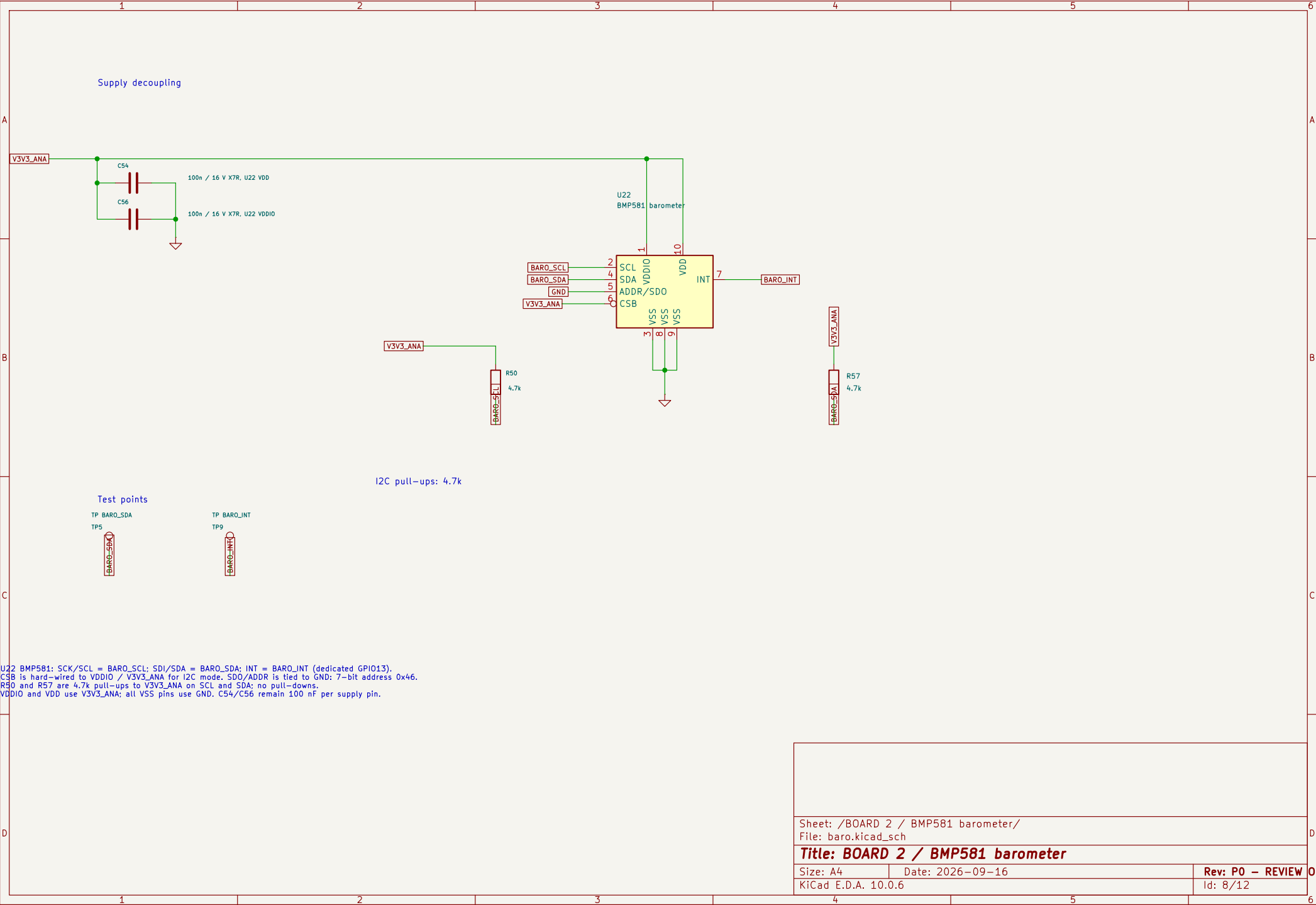
Dedicated buses: ADXL375 on hardware SPI0; ICM–45686 on hardware SPI1; BMP581 on hardware I2C0.
All sensors use V3V3_ANA. Existing supply decoupling is retained.
ICM–45686: C50/C52 are 100 nF X7R on VDD/VDDIO per datasheet Table 11.

Test points: TP1 ICM_SCK, TP2 ICM_MOSI, TP3 ICM_MISO, TP4 ICM_CS;
TP5 BARO_SDA, TP6 ADXL_CS; TP7 ICM_INT1, TP8 ICM_INT2.
TP9 BARO_INT, TP10 ADXL_INT1. Keep probe stubs short.
Routing priority: ICM SPI, ADXL SPI, native SD, then BMP581 I2C.
Use a continuous ground reference; avoid switching nodes and long runs beside servo/pyro power.

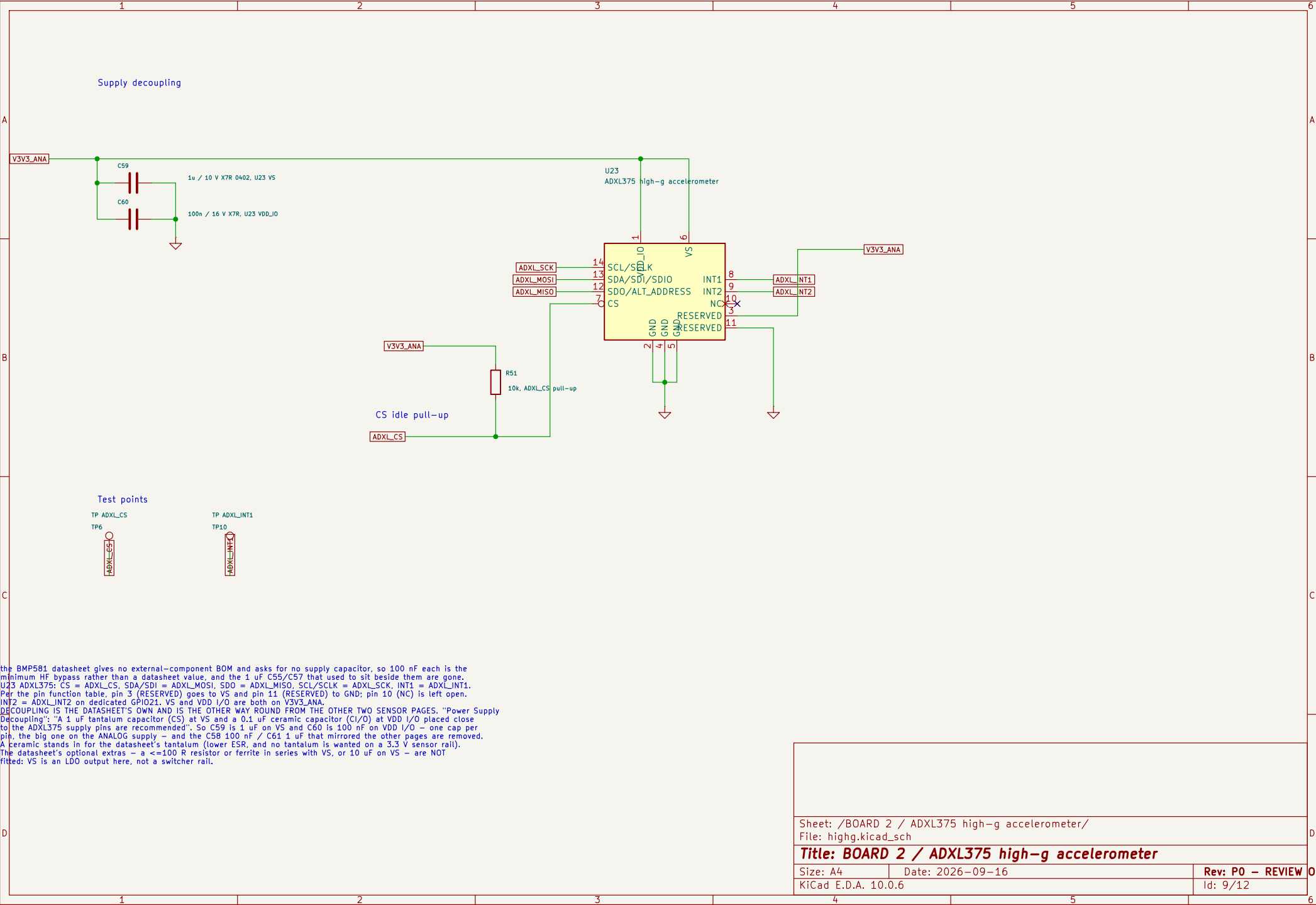
U21 ICM–45686 (TDK InvenSense DS–000577 Rev 1.0, Table 10 "Signal Descriptions" / Figure 4 pin-out, Single Interface SPI mode, Figure 10): AP_CS (pin 12) = ICM_CS, AP_SCLK (13) = ICM_SCK, AP_SDA/AP_SDIO/AP_SDI (14) = ICM_MOSI, AP_SDO/AP_ADO (1) = ICM_MISO, INT1 (4) = ICM_INT1, INT2/FSYNC/CLKIN (9) = ICM_INT2 – typed bidirectional because FSYNC/CLKIN are host–driven inputs while INT2 is a chip output; only the INT2 function is used here. VDD (8) and VDDIO (5) are both on V3V3_ANA; GND (6) is the only ground pin. Pins 2, 3, 7, 10 and 11 (RESV) are all left No Connect; Table 10 allows No Connect / VDDIO / GND for every RESV pin, and the Single–Interface SPI typical application schematic (Figure 10) shows all five left unterminated (internal pull enabled by default, per the per–pin notes in Table 10); they are only needed for the AUX1 or I2C–master dual–interface modes, neither of which this board uses. C50 (VDD) and C52 (VDDIO) are the 100 nF X7R of Table 11 "Bill of Materials" (C1/C2 in the datasheet) and are the WHOLE decoupling of this part: the BOM lists 0.1 uF per supply pin and nothing else, so the 1 uF C51/C53 this board used to add on top of them are removed. Table 11's own note – "use a larger bypass capacitor than 0.1 uF if power supply ripple exceeds 50 mV peak–to–peak" – is the condition that would bring them back; V3V3_ANA is an LDO output, not a switcher.
R48: 10k from ICM_CS to V3V3_ANA; R51: 10k from ADXL_CS to V3V3_ANA.
CS pull–ups hold each sensor deselected during reset. No pulls on SPI clock/data.
ICM_INT1 and ICM_INT2 each have their own MCU GPIO, independent of all other sensors.

			D
Sheet: /BOARD 2 / ICM-45686 six-axis IMU/ File: imu.kicad_sch			
Title: BOARD 2 / ICM-45686 six-axis IMU			
Size: A4	Date: 2026-09-16	Rev: P0 – REVIEW ONLY	6
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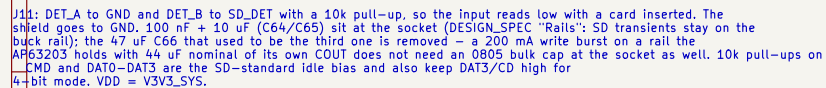
BOARD 2 – U22 BMP581 BAROMETER: hardware I2C0, powered from V3V3_ANA.



BOARD 2 – U23 ADXL375: dedicated hardware SPI0 (4–wire). Powered from V3V3_ANA.



VDD = V3V3_SYS¹; 10k pull-ups on CMD, DAT0-DAT3 and card detect. No clock pull.³



Sheet: /BOARD 3 / microSD socket/
File: sd.kicad_sch

Size: A4

Date: 2026-09-16

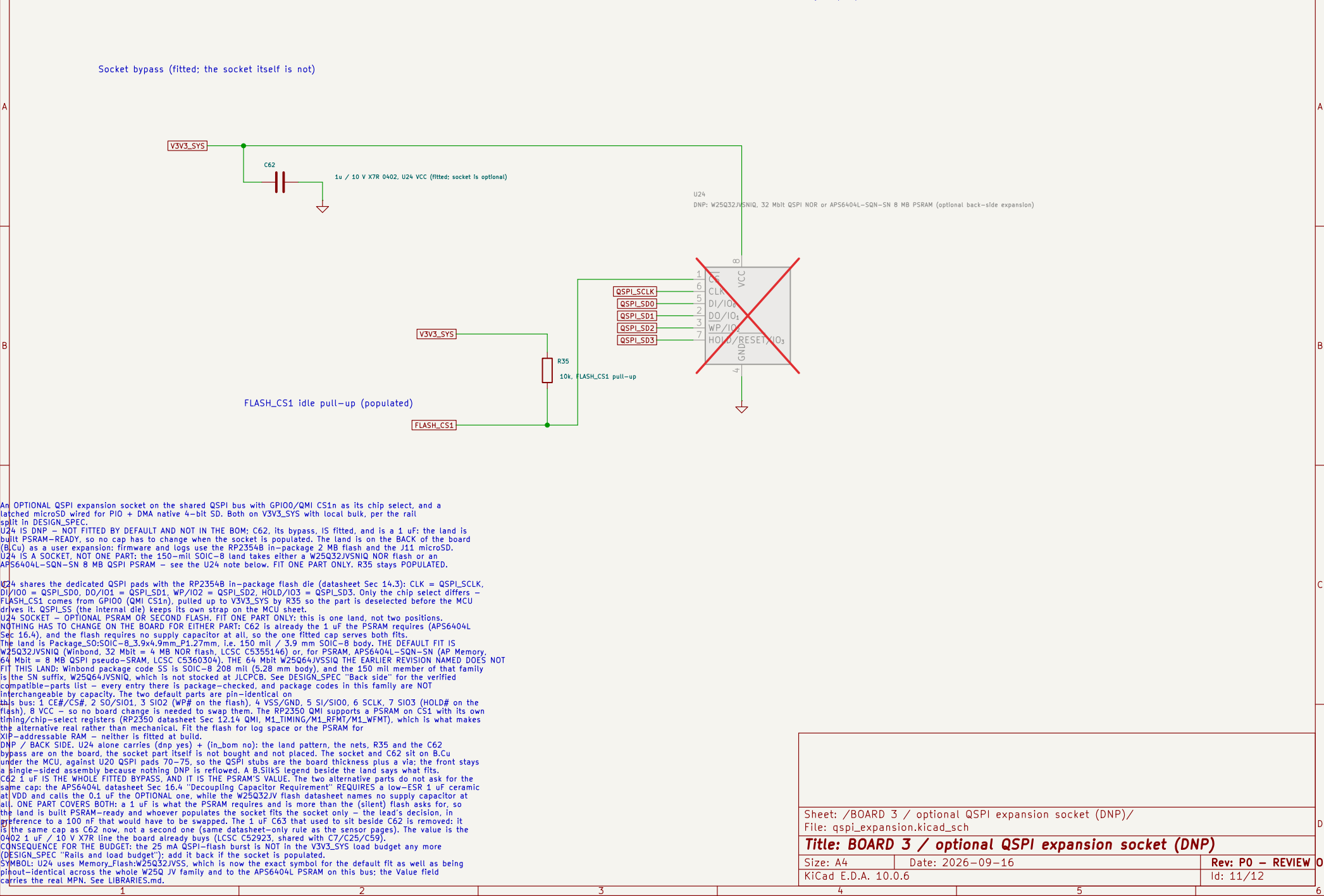
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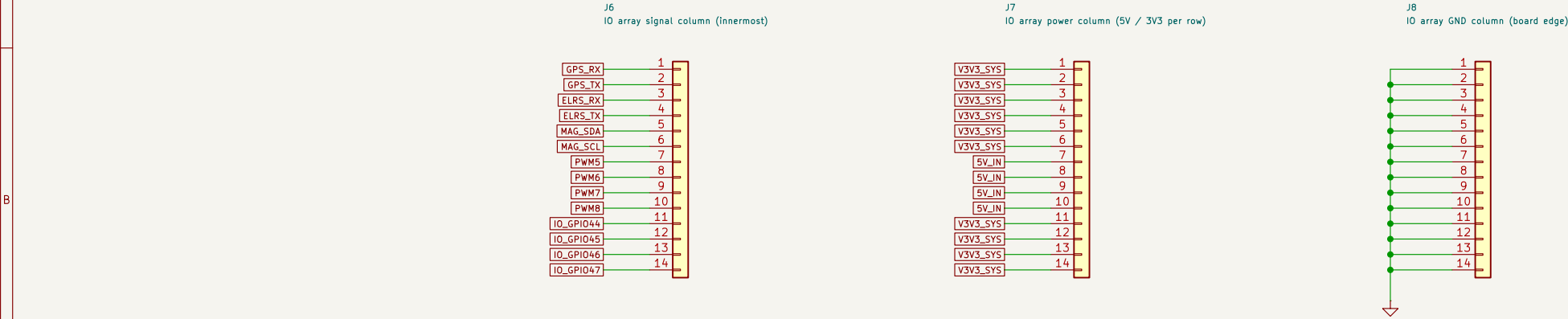
BOARD 3 – U24 OPTIONAL QSPI EXPANSION SOCKET, DNP, ON THE BACK (B.Cu).

One 150-mil SOIC-8 land: fit EITHER a W25Q32JVSNQ NOR flash OR an APS6404L-SQN-SN PSRAM. R35 stays populated.

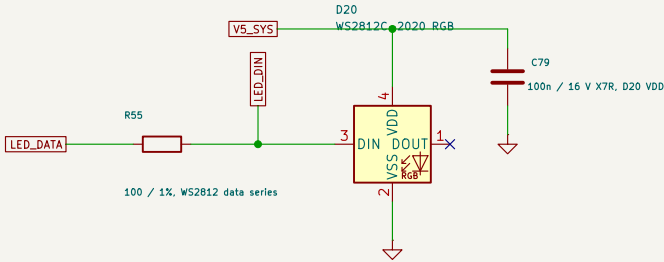


BOARD 4 – THE 3 x 14 IO BLOCK (J6 SIGNAL / J7 POWER / J8 GND), THE WS2812C STATUS LED AND THE FOUR M3 GROMMET HOLES. Every signal that leaves the board except the ESC row J3 and the DBG pads J10 leaves here.

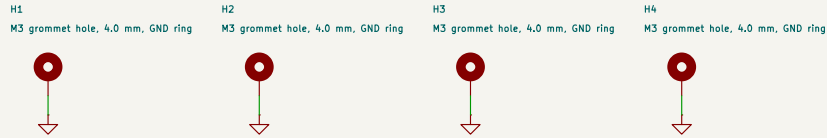
IO array – one row per module lead



WS2812C status LED



Mechanical



IO ARRAY – ONE 3–COLUMN x 14–ROW THROUGH–HOLE BLOCK ON THE RIGHT EDGE, 2.54 mm, madflight FC3v2 style. It replaces the old 2x16 left–edge array AND the J12/J13/J14 servo block: every signal that leaves this board except the ESC row (J3) and the SWD pads (J10) now leaves on this one block. COLUMN RULE, from the board edge inward: GND (J8) | POWER (J7) | SIGNAL (J6). The SIGNAL column is the INNERMOST one, next to the MCU, so the 14 signal traces are the short ones; the two rails run down the outside where a plane stitch is cheap. A 3–wire module lead (signal, power, ground) plugs straight across ONE row. THREE Conn_01x14 SYMBOLS, not one 3x14: no 3–row generic connector symbol ships with this KiCad install, and three 1x14 rows with a body–trimmed courtyard about exactly on the 2.54 mm grid (see tools/setup_pcb.py). DS–009 NAMING IS KEPT: a port label is named for the PERIPHERAL pin it belongs to, so GPS_RX / ELRS_RX are module inputs driven by the RP2354B UART TX pins (rows 1 and 3, silk T0 / T1) and GPS_TX / ELRS_TX are module outputs that land on the MCU UART RX pins (rows 2 and 4, silk R0 / R1).

PWM2 = GPIO39
PWM3 = GPIO38
PWM4 = GPIO36

RAILS ON THE ARRAY: rows 1–6 (GPS, ELRS, magnetometer) and rows 11–14 take V3V3_SYS; rows 7–10 (servos S5–S8) take 5V_IN. Every row has its own GND pin in the J8 column, so a signal return is never more than 2.54 mm away and no row can be mis–plugged into a neighbour’s ground. THE FOUR 5 V SERVO ROWS ARE ON 5V_IN, AHEAD OF THE OR–ING FET (DESIGN_SPEC “Decisions”, 2026–09–17). 5V_IN is the external BEC on J3 pin 7, so servo and module current goes from the BEC straight to the rows and never crosses Q1: the FET carries only the two 3.3 V regulators and the status LED, and a USB host never sees the servo rows at all. The rows are therefore NOT powered from USB – plugging in USB alone leaves every module and servo dead, which is the intended bench behaviour and the reverse of the previous revision. CURRENT: rows 7–10 count against the external BEC; rows 1–6 and 11–14 count in the V3V3_SYS budget. No fuse and no per–row current limit: the array is a system–integration connector, not a protected port. NO LOCAL BULK AT THE ARRAY ANY MORE: C20/C21 (10 uF on 5V_IN) and C22 (10 uF on V3V3_SYS) are removed. The 5 V rows are fed by the external BEC through C19 100 uF at the J3 pad row, and the 3.3 V rows by the AP63203 and its 44 uF nominal of COUT: three 0603 at the connector added nothing either rail did not already have, and a module’s own inrush is its own problem – this block is a connector, not a supply.

MECHANICAL: H1–H4 are the four 4.0 mm grommet holes on the standard 30.5 x 30.5 mm pattern, off–centre in X, each now a PLATED hole with a 6.4 mm GND pad ring stitched by eight 0.6/0.3 mm vias, and a 6.5 mm top–side courtyard for the grommet flange. Pin 1 of every hole is GND, so the frame screws bond to the ground plane on every layer. See DESIGN_SPEC “Mounting and orientation”.

IO ARRAY ROW MAP (row n = J6 pin n signal J7 pin n power J8 pin n GND)				
1	T0	GPS_RX	V3V3_SYS	GND UART1 TX -> GPS RX (GPIO24)
2	R0	GPS_TX	V3V3_SYS	GND UART1 RX <- GPS TX (GPIO25)
3	T1	ELRS_RX	V3V3_SYS	GND UART0 TX -> ELRS RX (GPIO16)
4	R1	ELRS_TX	V3V3_SYS	GND UART0 RX <- ELRS TX (GPIO17)
5	SDA	MAG_SDA	V3V3_SYS	GND I2C1 SDA (GPIO22), 4.7k pull–up on the MCU sheet
6	SCL	MAG_SCL	V3V3_SYS	GND I2C1 SCL (GPIO23), 4.7k pull–up on the MCU sheet
7	S5	PWM5	5V_IN	GND PWM2 A (GPIO20), servo 5
8	S6	PWM6	5V_IN	GND PWM1 B (GPIO19), servo 6
9	S7	PWM7	5V_IN	GND PWM7 B (GPIO15), servo 7
10	S8	PWM8	5V_IN	GND PWM2 B (GPIO5), servo 8
11	A44	IO_GPIO44	V3V3_SYS	GND GPIO44 = ADC4, spare
12	A45	IO_GPIO45	V3V3_SYS	GND GPIO45 = ADC5, spare
13	A46	IO_GPIO46	V3V3_SYS	GND GPIO46 = ADC6, spare
14	A47	IO_GPIO47	V3V3_SYS	GND GPIO47 = ADC7, spare

PWM MAP
PWM5 = GPIO20 (row 7, S5)
PWM6 = GPIO19 (row 8, S6)
PWM7 = GPIO15 (row 9, S7)
PWM8 = GPIO5 (row 10, S8)

ON THE ESC PAD ROW (J3, POWER 1)
PWM1 = GPIO43

Sheet: /BOARD 4 / IO array, status LED, mounting holes/
File: io_block.kicad_sch

Title: BOARD 4 / IO array, status LED, mounting holes

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